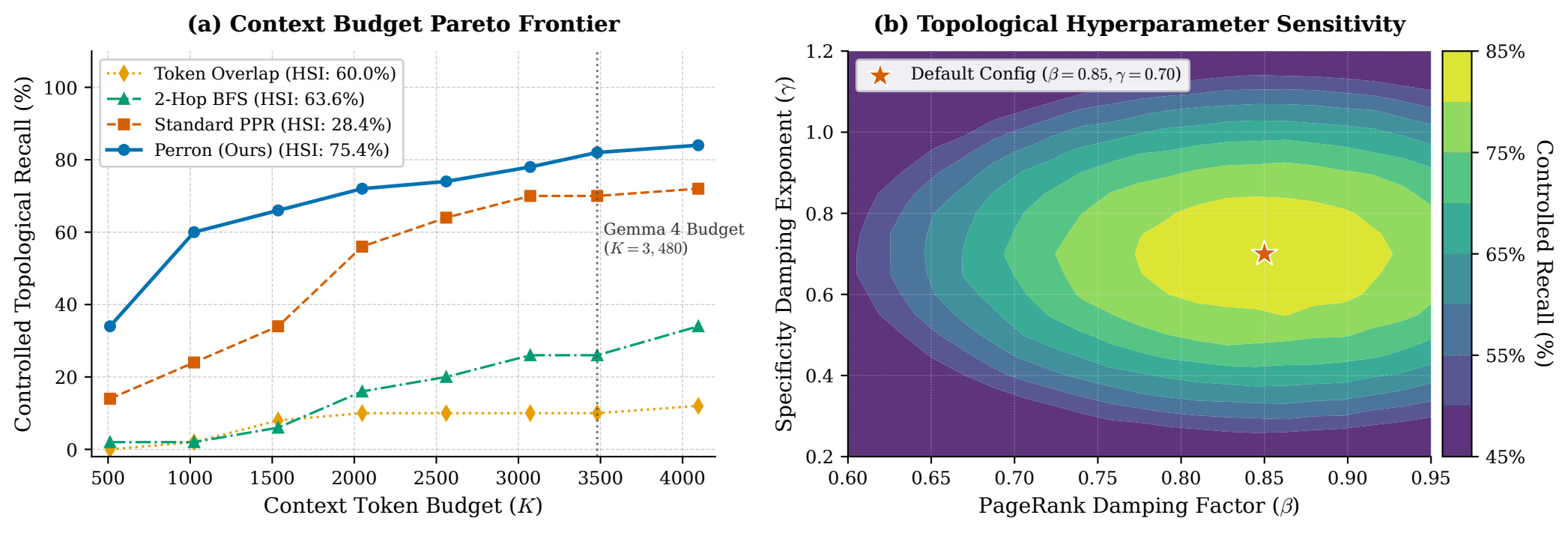
(a) Context Budget Pareto Frontier
Token Overlap (HSI: 60.0%)
2-Hop BFS (HSI: 63.6%)
Standard PPR (HSI: 28.4%)
Perron (Ours) (HSI: 75.4%)
Gemma 4 Budget
($K = 3,480$)
Controlled Topological Recall (%)
Context Token Budget (K)
0
20
40
60
80
100
500
1000
1500
2000
2500
3000
3500
4000
(b) Topological Hyperparameter Sensitivity
Default Config ($\beta = 0.85, \gamma = 0.70$)
Specificity Damping Exponent (γ)
PageRank Damping Factor (β)
Controlled Recall (%)
0.2
0.4
0.6
0.8
1.0
1.2
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
45%
55%
65%
75%
85%